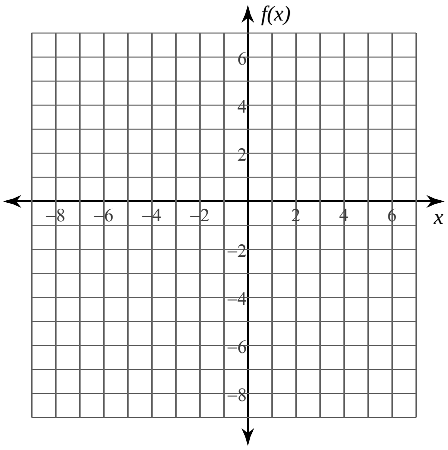
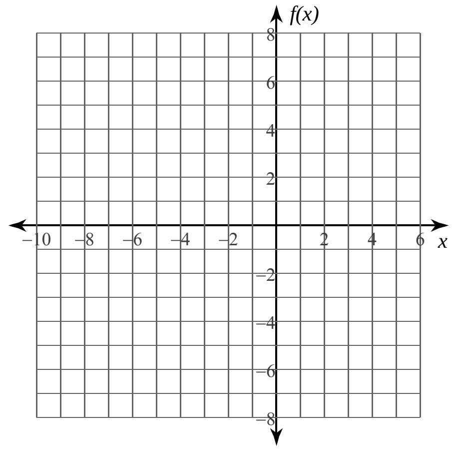
Evaluate each limit. You may use the provided graph to sketch the function.

5) $\lim\limits_{x \to -1^-} f(x),\ f(x) = \begin{cases} x + 2, & x \le -1 \\ -\dfrac{x}{2} - 4, & x > -1 \end{cases}$



6) $\lim\limits_{x \to -2} f(x),\ f(x) = \begin{cases} x^2 + 6x + 8, & x < -2 \\ -\dfrac{x}{2} - 1, & x \ge -2 \end{cases}$



Evaluate each limit.

7) $\lim\limits_{x \to 4^-} f(x),\ f(x) = \begin{cases} 2x - 5, & x < 4 \\ -2x + 7, & x \ge 4 \end{cases}$

8) $\lim\limits_{x \to 3^-} f(x),\ f(x) = \begin{cases} -x^2 + 4x - 3, & x < 3 \\ \dfrac{x}{2} - 3, & x \ge 3 \end{cases}$

9) $\lim\limits_{x \to 1^+} f(x),\ f(x) = \begin{cases} 2x - 5, & x < 1 \\ x - 4, & x \ge 1 \end{cases}$

10) $\lim\limits_{x \to -3^-} f(x),\ f(x) = \begin{cases} -x^2 - 4x - 4, & x < -3 \\ -1, & x \ge -3 \end{cases}$

Critical thinking question:

11) Write a piecewise function where $\lim\limits_{x \to 2^-} f(x) = 1$, $\lim\limits_{x \to 2^+} f(x) = 3$, and $f(2) = 2$.